

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)**UNIT TEST****29-11-2020****JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]**

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on this Test Booklet.

PAPER-2

Paper : Physics, Chemistry & Mathematics

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **25 questions** in each subject and each subject having **Two sections**.
 - (i) Section-I contains 20 multiple choice questions with **only one correct** option.
Marking scheme : +4 for correct answer, 0 if not attempted and -1 in all other cases.
 - (ii) Section-II contains 5 **Numerical Value Type** questions
Marking scheme : +4 for correct answer and 0 in all other cases.
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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PART 1 - PHYSICS

SECTION-I : (Maximum Marks : 80)

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

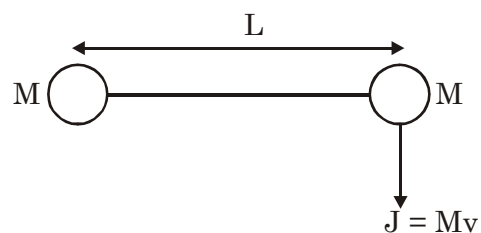
Negative Marks : -1 In all other cases

1. An object has velocity $\vec{v}_1$ relative to the ground. An observer moving past with a constant velocity $\vec{v}_0$ relative to the ground measures the velocity of the object to be $\vec{v}_2$ (relative to the observer). The magnitudes of these velocities are related by :-
- (A) $v_0 \leq v_1 + v_2$
 (B) $v_1 \leq v_2 + v_0$
 (C) $v_2 \leq v_0 + v_1$
 (D) All the above are true

2. A motorcycle moves around a vertical circle with a constant speed under the influence of the force of gravity $\vec{W}$, the force of friction between the wheels and the track $\vec{f}$, and the normal force between the wheels and the track $\vec{N}$. Which of the following quantities has a constant magnitude ?

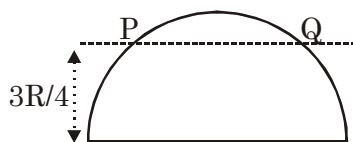
- (A) $\vec{N}$ (B) $\vec{N} + \vec{f}$
 (C) $\vec{f} + \vec{W}$ (D) $\vec{N} + \vec{W} + \vec{f}$

3. Consider a body, shown in figure, consisting of two identical balls, each of mass M connected by a light rigid rod. If an impulse $J = Mv$ is imparted to the body at one of its end, what would be its angular velocity ?



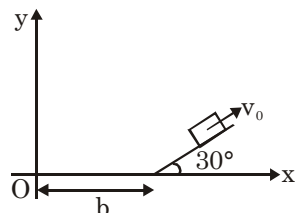
- (A) $\frac{v}{L}$ (B) $\frac{2v}{L}$
 (C) $\frac{v}{3L}$ (D) $\frac{v}{4L}$

4. The radius of gyration of a solid hemisphere of mass M and radius R about an axis parallel to the diameter at a distance $\frac{3}{4}R$ is given by (centre of mass of the hemisphere lies at a height $\frac{3R}{8}$ from the base) :-



- (A) $\frac{3R}{\sqrt{10}}$ (B) $\frac{5R}{4}$
 (C) $\frac{5R}{8}$ (D) $\sqrt{\frac{2}{5}}R$
5. A cube of mass m and side a is moving along a plane with constant speed v_0 as shown in figure. The magnitude of angular momentum of the cube about z -axis would be :-

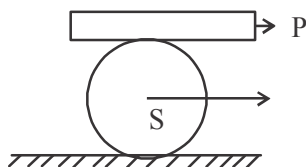
- (A) $\frac{mv_0 b}{2}$
 (B) $\frac{\sqrt{3}mv_0 b}{2}$



- (C) $mv_0 \left(b - \frac{a}{2} \right)$
 (D) None of these

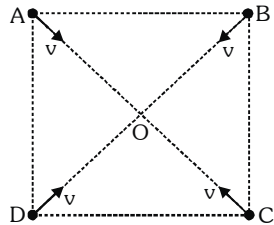
6. A basket ball player jumps up to "shoot" a basket. Is momentum conserved ?
 (A) Yes, but only if you choose the correct system
 (B) Yes, but only in the horizontal direction
 (C) No, because the velocity of the basketball player changes with time.
 (D) It is a bad question, because momentum conservation is for objects moving at constant speed, and the basketball player is accelerating.
7. Two objects of unequal mass are connected by a light string that passes over a pulley. One of the object is given an initial upward velocity. The center of mass of the two objects will :-
 (A) Accelerate up or down, depending on the relative masses of the two objects.
 (B) Accelerate downward only after it has reached some highest point.
 (C) Accelerate downward at some value less than g .
 (D) Accelerate downward with a value of g .
8. Two different disks of radii $r_1 > r_2$ are free to spin separately about an axis through the center and perpendicular to the plane of each disk. Both disks start from rest, and both undergo the same angular acceleration for the same length of time. Which disk will have the larger final angular velocity ?
 (A) Disk 1
 (B) Disk 2
 (C) The disks will have the same angular velocity
 (D) The answer depends on the mass of the disks.

9. Which way does $\vec{\omega}$ point for the Earth ?
 (A) Parallel to the NS axis and pointing north
 (B) Parallel to the NS axis and pointing south
 (C) Parallel to the NS axis and pointing east.
 (D) Parallel to the NS axis and pointing west
10. A body is thrown with a velocity of 10 m/s at an angle of 45° to the horizontal. Acceleration of free fall to be 10 m/s^2 . The radius of curvature of its trajectory $1/\sqrt{2}$ seconds after the body begin to move is :
 (A) 0 m (B) 2.5m
 (C) 5m (D) None
11. A plank 'P' is placed on a solid cylinder S which rolls on a horizontal surface. The two are of equal mass. There is no slipping at any of the surfaces in contact. The ratio of kinetic energy of 'P' to the kinetic energy of 'S' is :-



- (A) 1:1 (B) 12:1
 (C) 8 :3 (D) 11:8

12. An engineer wants to design an improved elevator for a building. The original design used a motor that could lift 1000 kg through a distance of 20 meters in 30 seconds. The engineer wants a motor that can lift 800 kg through a distance of 30 meters in 20 seconds. Compared to the old motor, the new motor :-
 (A) Should exert a force of the same magnitude, but must provide a larger power output.
 (B) Should exert a force of a larger magnitude, and must provide a larger power output
 (C) Can exert a force of a smaller magnitude, and can provide a smaller power output.
 (D) Can exert a force of a smaller magnitude, but must provide a larger power output.
13. Seven identical geese are flying south together at constant speed. A hunter shoots one of them, which immediately dies and falls to the ground. The other six continue flying south at the original speed. After the one goose has hit the ground, the center of mass of the all seven geese :-
 (A) Continues south at the original speed, but is now located some distance behind the flying geese.
 (B) Continues south, but at $\frac{6}{7}$ the original speed.
 (C) Continues south, but at $\frac{1}{7}$ the original speed.
 (D) Stops with the dead goose.

14. A ladder is at rest with its upper end against a wall and its lower end on the ground. A worker is about to climb it. When is it more likely to slip ?
 (A) Before the worker is on it.
 (B) When the worker is on the lowest rung.
 (C) When the worker is halfway up the ladder.
 (D) When the worker is on the top rung
15. A particle is in a unidirectional potential field where the potential energy (U) of a particle depends on the x -coordinate given by $U_x = a(1 - \cos bx)$ & a and b are constants. The physical units of ' b '
 (A) $\text{kg}^0\text{m}^0\text{sec}^1$ (B) $\text{kg}^0\text{m}^{-1}\text{sec}^0$
 (C) $\text{kg}^1\text{m}^0\text{sec}^1$ (D) $\text{kg}^1\text{m}^1\text{sec}^1$
16. A block slides from rest down a wedge inclined at an angle θ with the horizontal. There is friction between the block and the wedge. When the block reaches the bottom of the wedge, its kinetic energy is 3J , and gravity has done $+10\text{J}$ of work on the block. Which of the following describes the energy transfers in this system ?
 (A) $\Delta E_{\text{int, block}} < +7\text{ J}$
 (B) Frictional work on block by wedge $= -7\text{J}$
 (C) Frictional work on wedge by block $= +7\text{ J}$
 (D) $\Delta E_{\text{int, block}} = +7\text{ J}$
17. A particle with total energy E moves in one dimension in a region where the potential energy is $U(x)$. The speed of the particle is zero where
 (A) $U(x) = E$ (B) $U(x) = 0$
 (C) $\frac{dU(x)}{dx} = 0$ (D) $\frac{d^2U(x)}{dx^2} = 0$
18. Four identical particles A, B, C and D move with equal speed v , along the diagonal of a square towards the centroid O of the square, beginning at the vertices. After the collision at O, A and C retrace their path with speed $2v$ and B retraces its path with speed $v\sqrt{2}$. Speed of the particle D after the collision is
- 
- (A) $\sqrt{2}v$ (B) v (C) $\frac{v}{\sqrt{2}}$ (D) $2v$
19. A spherical constant temperature heat source of radius r_1 is at the center of a uniform solid sphere of radius r_2 . The rate at which heat is transferred through the surface of the sphere is proportional to :-
 (A) $\left(\frac{1}{r_2} - \frac{1}{r_1}\right)^{-1}$ (B) $r_2 - r_1$
 (C) $\ln r_1 - \ln r_2$ (D) $\frac{1}{r_2} - \frac{1}{r_1}$
20. A uniform rod AB of mass m and length $2a$ is falling freely without rotation under gravity with AB horizontal. Suddenly the end A is fixed when the speed of the rod is v . The angular speed with which the rod begins to rotate is :-
 (A) $\frac{v}{2a}$ (B) $\frac{4v}{3a}$ (C) $\frac{v}{3a}$ (D) $\frac{3v}{4a}$

SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

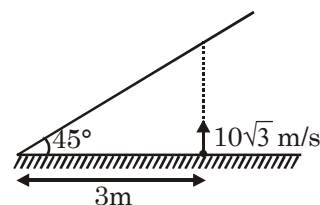
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

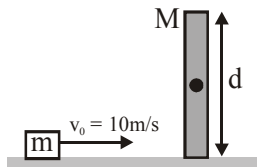
-									
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

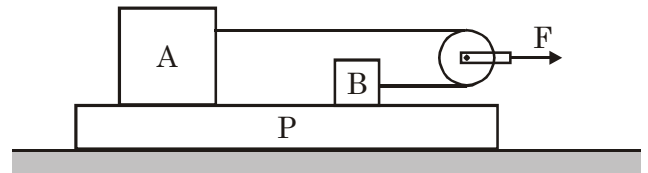
1. An elastic light cord of length 10m is taut between two nails that are in the same level. A load suspended at the midpoint of the cord, if allowed to move gradually, stops after descending a height $y_1 = 1$ cm. If the same load is dropped from a height above the midpoint of the cord, the load on striking the midpoint sticks there and then descends a maximum height $y_2 = 2$ cm. Assuming collision of the load with the cord to be lossless, estimate the height (in cm) above the cord from where the load was dropped.
2. A particle is thrown upward with speed $10\sqrt{3}$ m/s. It strikes the inclined surface as shown in figure. If collision is perfectly inelastic, find maximum height of particle from ground ($g = 10$ m/s²)



3. A cube of mass m slides without friction at speed $v_0 = 10 \text{ m/s}$. It undergoes a perfectly elastic collision with the bottom tip of a rod of length d and mass $M = 2m$. The rod is pivoted about a frictionless axle through its center, and initially it hangs straight down and is at rest. What is the cube's velocity (In m/s) after collision.



4. In the setup shown, blocks A and B of masses 4.0 kg and 1.0 kg are placed on a platform P of mass 1.0 kg that is placed on a horizontal frictionless floor. The blocks are connected by a light inextensible string that passes round an ideal pulley. Coefficients of static and kinetic friction between the blocks and the platform are 0.16 and 0.10 . On applying a horizontal force F on the pulley, acceleration of the platform becomes 2.0 m/s^2 . Find magnitude of the force F (in N).



5. Kinetic energy of a particle moving on a circle is given as, $\text{K.E.} = \frac{1}{2} m^a \omega^b r^c$, where m = mass, ω = angular speed, r = radius of circle then, $|a + b + c|$ would be :

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PART 2 - CHEMISTRY

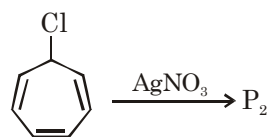
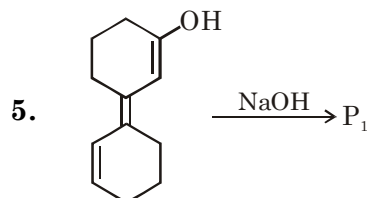
SECTION-I : (Maximum Marks : 80)

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
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Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Which of the following oxyacid does not contain X-X linkage? (where X = S/P)
 (A) $\text{H}_2\text{S}_2\text{O}_5$ (B) $\text{H}_2\text{S}_2\text{O}_6$
 (C) $\text{H}_4\text{P}_2\text{O}_7$ (D) $\text{H}_2\text{S}_2\text{O}_4$
2. Which of the following is/are not dicarboxylic acid :
 (A) Succinic acid (B) Oxalic acid
 (C) Acrylic acid (D) Malonic acid
3. A 50 ml portion of mixture H_2SO_4 & $\text{H}_2\text{C}_2\text{O}_4$ required 200 ml 0.10 M $\text{Ba}(\text{OH})_2$ solution for titration. Another 50 ml required 200ml , 0.1M KMnO_4 solution for titration in strong basic medium. Calculate difference in mass of H_2SO_4 and $\text{H}_2\text{C}_2\text{O}_4$ in mixture
 (A) 0.08 gm (B) 0.98 gm
 (C) 0.8 gm (D) 0.9 gm

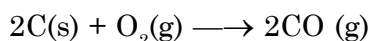
4. Select the **CORRECT** order according to given property?

- (A) $\text{N}_2\text{H}_4 < \text{N}_2\text{F}_4$ (N-N bond length)
 (B) $\text{CH}_3\text{F} < \text{CH}_2\text{F}_2$ (C-F bond length)
 (C) $\text{SOCl}_2 < \text{SOF}_2$ (S-X bond length)
 (D) $\text{CF}_3\text{Cl} < \text{CH}_3\text{Cl}$ (C-Cl bond length)



If number of resonating structure for P_1 and P_2 are X & Y respectively. Then (Y-X) has value.

- (A) 6 (B) 4 (C) 3 (D) Zero
6. Certain amount of oxygen gas was passed over heated carbon, where 80% of oxygen added was converted into CO gas, as per reaction



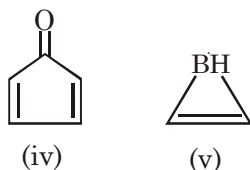
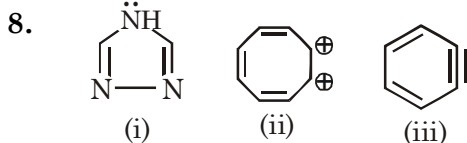
If gaseous mixture left is allowed to effuse then ratio of rate of effusion of CO to O_2 will be

- (A) $\sqrt{\frac{8}{7}}$ (B) $\sqrt{\frac{8^3}{7}}$ (C) $\sqrt{\frac{4^2 \times 8}{7}}$ (D) $\sqrt{\frac{2^2 \times 8}{7}}$

7. Select the **CORRECT** match?

Silicate unit	Number of unshared oxygen corner in per tetrahedral unit
---------------	--

- | | |
|---|---|
| (A) $(\text{SiO}_3^{-2})_n$ | 2 |
| (B) $(\text{Si}_4\text{O}_{11}^{-6})_n$ | 2 |
| (C) $\text{Si}_3\text{O}_9^{-6}$ | 3 |
| (D) $(\text{SiO}_2)_n$ | 3 |



Correct set of aromatic species are :

- | | |
|-------------------|------------------|
| (A) i, iii, iv | (B) ii, iv, v |
| (C) i, ii, iii, v | (D) All of these |

9. Which relationship does hold for vander-wall gas under all conditions, where all terms have their usual meaning.

(A) $PV = nRT$

(B) $\left(P + \frac{a}{n^2V^2}\right)(V - nb) = nRT$

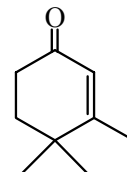
(C) $\left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$

(D) $\left(P + \frac{a^2n}{V^2}\right)(V - nb) = nRT$

10. Which of the following order is **CORRECT** for mentioned property but this order is **NOT** explained by hydrogen bonding -

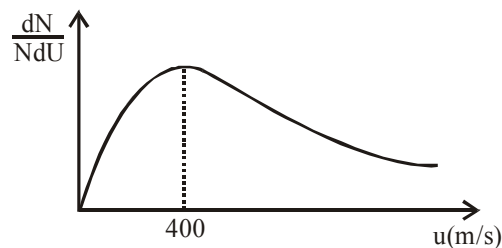
- (A) Melting point : $\text{NH}_3 > \text{SbH}_3$
 (B) Density of H_2O : Liquid > Solid
 (C) Boiling point : $\text{H}_2 > \text{He}$
 (D) Boiling point : $\text{H}_2\text{O} > \text{H}_2\text{Te}$

11. How many hydrogen will be replaced by deuterium when following compound kept in $\text{NaOD}/\text{D}_2\text{O}$ solution for long time.



- (A) 3 (B) 4 (C) 5 (D) 6

12. For ideal gas following observation is made as per Maxwell-Boltzman distribution, select the **INCORRECT** option :



(A) $u_{\text{rms}} = \sqrt{1.5} \times 400 \text{ m/s}$

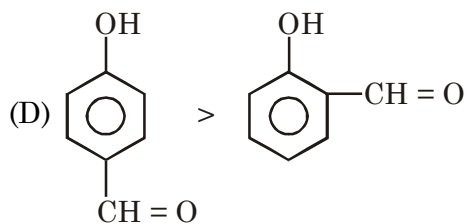
(B) $u_{\text{Avg.}} = \sqrt{\frac{4}{\pi}} \times 400 \text{ m/s}$

(C) Fraction of molecules moving between 400 to 401 m/s are more than fraction of molecules moving between 452 to 453 m/s

(D) Fraction of molecules moving between 400 to 401 m/s are equal with fraction of molecules moving between 452 to 453 m/s

13. **INCORRECT** order of boiling point?

- (A) $\text{Xe} > \text{Kr} > \text{Ar} > \text{Ne} > \text{He}$
 (B) n-pentane > Iso pentane > neo pentane
 (C) $\text{SiCl}_4 > \text{CCl}_4$



14. Compound in which one of the Tautomeric form can show Geometrical isomerism is :

- (A) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$ (B) $\text{CH}_3-\text{CH}_2-\text{NO}_2$



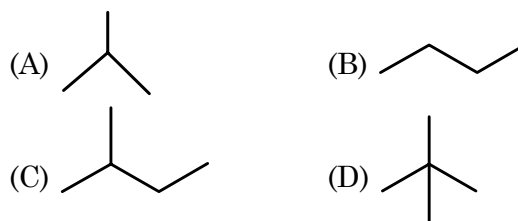
15. A photon of wavelength 155 nm is fallen on the metal surface (work function = $64 \times 10^{-20} \text{ J}$). The electron ejected from the metal surface with maximum kinetic energy is absorbed by Hg vapours and emitted radiation by losing its kinetic energy. The wavelength of emitted radiation is:

- (A) 155 nm (B) 165 nm
 (C) 310 nm (D) 620 nm

16. Which of the following combination of orbitals not able to form π -bond through any axis of approach?

- (A) p_z and d_{xz} (B) p_z and p_z
 (C) p_z and d_{z^2} (D) p_y and p_y

17. How many compounds are structural isomer of iso pentane



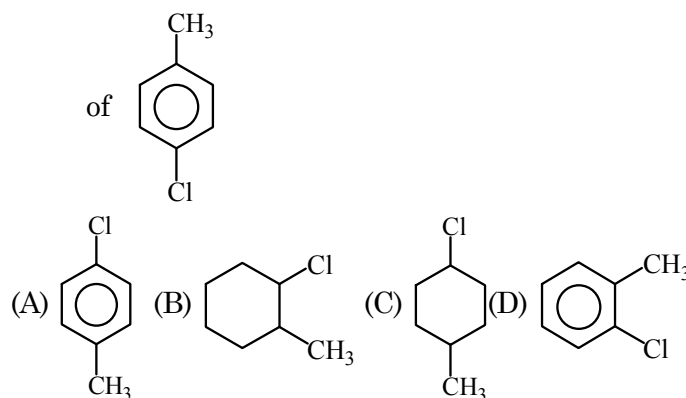
18. Which transition of electron in Li^{2+} would emit the photon whose wavelength will be same as that of photon emitted by Hydrogen atoms corresponding to 2nd line of Balmer series.

- (A) $12 \rightarrow 6$ (B) $8 \rightarrow 4$
 (C) $8 \rightarrow 2$ (D) $6 \rightarrow 3$

19. Which of the following species have different electron geometry and shape?

- (A) XeO_3F_2 (B) IF_7
 (C) SiF_6^{2-} (D) XeF_5^+

20. Which of the following is positional isomer



SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

⊕	●				
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	●	7	7
8	8	8	8	8	8
9	9	9	9	9	9

●	⊖				
●	●	0	0	●	●
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	●	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Calculate molality of one litre of solution containing 200 gm CaBr_2 . Given density of solution equal to 1.0 gm/ml.
(Atomic mass of Ca = 40, Br = 80)
2. Find the total number of species which do not exist.
 BrH_5 , BF_6^{-3} , PI_6^- , SiCl_6^{-2} , ICl_7 , ClF_7 , SI_6 , IH_3 , HFO_4
3. Total number structure of ester possible with molecular formula $\text{C}_5\text{H}_{10}\text{O}_2$
4. A gaseous hydrocarbon (C_xH_y) requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times of its volume of CO_2 . Find out the volume of x + y.
5. Total number of polar molecule/s in which both polar and non polar bond is present.
 PCl_3F_2 , H_2O_2 , CCl_4 , O_2F_2 , C_2H_4 , S_2F_{10}

ALLEN
PART 3 - MATHEMATICS

SECTION-I : (Maximum Marks : 80)

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +4 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Given $A \equiv (1, 1)$ and AB is any line through it cutting the x-axis at B. If AC is perpendicular to AB and meets the y-axis in C, then the equation of the locus of midpoint P of BC is
 (A) $x + y = 1$ (B) $x + y = 2$
 (C) $x + y = 2xy$ (D) $2x + 2y = 1$
2. The value of $\sin^3 10^\circ + \sin^3 50^\circ - \sin^3 70^\circ$ is equal to
 (A) $-\frac{3}{2}$ (B) $\frac{3}{4}$ (C) $-\frac{3}{4}$ (D) $-\frac{3}{8}$
3. The angle between the tangents to the parabola $y^2 = 4ax$ at the points where it intersects with the line $x - y - a = 0$ is
 (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

4. If A, B, C are the angles of a triangle such that $\cot \frac{A}{2} = 3 \tan \frac{C}{2}$, then $\sin A, \sin B, \sin C$ are in
 (A) A.P
 (B) G.P
 (C) H.P
 (D) none of these
5. A tangent to the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ at any point P meets the line $x = 0$ at a point Q. Let R be the image of Q in the line $y = x$. then the circle whose extremities of a diameter are Q and R passes through a fixed point. The fixed point is -
 (A) (3, 0) (B) (5, 0)
 (C) (0, 0) (D) (4, 0)
6. $\cos^2 \alpha + \cos^2(\alpha + 120^\circ) + \cos^2(\alpha - 120^\circ)$ is equal to
 (A) $\frac{3}{2}$ (B) 1
 (C) $\frac{1}{2}$ (D) 0
7. Three circles with centres at A, B, C intersect orthogonally. The point of intersection of the common chords is
 (A) orthocentre of $\triangle ABC$
 (B) circumcentre of $\triangle ABC$
 (C) incentre of $\triangle ABC$
 (D) centroid of $\triangle ABC$

8. The locus of the mid-points of the chords of the circles $x^2 + y^2 - 2x - 4y - 11 = 0$ which subtends an angle of 60° at centre is-
 (A) $x^2 + y^2 - 4x - 2y - 7 = 0$
 (B) $x^2 + y^2 + 4x + 2y - 7 = 0$
 (C) $x^2 + y^2 - 2x - 4y - 7 = 0$
 (D) $x^2 + y^2 + 2x + 4y + 7 = 0$
9. A ray of light incident at the point (3, 1) gets reflected from the tangent at (0, 1) to the circle $x^2 + y^2 = 1$. The reflected ray touches the circle. The equation of the line containing the reflected ray is
 (A) $3x + 4y - 13 = 0$ (B) $3x - 4y - 5 = 0$
 (C) $3x - 4y + 13 = 0$ (D) $4x - 3y - 10 = 0$
10. If the normal at $(ap^2, 2ap)$ on the parabola $y^2 = 4ax$, meets the parabola again at $(aq^2, 2aq)$, then
 (A) $p^2 + pq + 2 = 0$ (B) $p^2 - pq + 2 = 0$
 (C) $q^2 + pq + 2 = 0$ (D) $p^2 + pq + 1 = 0$
11. P(3,2) is a point on the circle $x^2 + y^2 = 13$. Two points A, B are on the circle such that $PA = PB = \sqrt{5}$. The equation of chord AB is
 (A) $4x - 6y + 21 = 0$ (B) $6x + 4y - 21 = 0$
 (C) $4x + 6y - 21 = 0$ (D) $6x + 4y + 21 = 0$
12. If 19 arithmetic means $A_1, A_2, A_3, \dots, A_{19}$ are inserted between 102 and 2 such that their sum is S, then the value of $\frac{S}{(A_{17} + 2)A_{10}}$ is
 (A) 1 (B) 17 (C) 52 (D) 5
13. The length of side of an equilateral triangle with one vertex at origin and inscribed in $y^2 = 16x$ is-
 (A) $4\sqrt{3}$ (B) $16\sqrt{3}$ (C) $32\sqrt{3}$ (D) $24\sqrt{3}$
14. S and T are the foci of an ellipse and B is an end of the minor axis. If STB is an equilateral triangle, the eccentricity of the ellipse is
 (A) $1/4$ (B) $1/3$ (C) $1/2$ (D) $2/3$
15. The locus of the center of the circle touching the line $2x - y = 1$ at (1, 1) is-
 (A) $x + 3y = 2$ (B) $x + 2y = 0$
 (C) $x + y = 2$ (D) None of these
16. $\sqrt{x+3-4\sqrt{x-1}} + \sqrt{x+8-6\sqrt{x-1}} = 1$.
 Solve for x
 (A) {5, 10} (B) [1, ∞)
 (C) [5, 10] (D) none of these
17. The greatest value of $\sin^4\theta + \cos^4\theta$ is
 (A) $\frac{1}{2}$ (B) 1 (C) 2 (D) 3
18. If $\sin\alpha, \sin\beta$ and $\cos\alpha$ are in G.P, then roots of the equation $x^2 + 2x\cot\beta + 1 = 0$ are always.
 (A) equal (B) real
 (C) imaginary (D) greater than 1
19. The number of solutions of $\sin^3x\cos x + \sin^2x\cos^2x + \sin x\cos^3x = 1$ in $[0, 2\pi]$ is
 (A) 4 (B) 2 (C) 1 (D) 0
20. $\Delta = \begin{vmatrix} x^2 + y^2 & ax + by & px + qy \\ ax + by & a^2 + b^2 & ap + bq \\ px + qy & ap + bq & p^2 + q^2 \end{vmatrix}$ is equal to
 (A) $p + q$
 (B) $a + b + c$
 (C) $x + y + z$
 (D) 0

SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	●	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	②	●	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If $2\sin x - \cos 2x = 1$, then $\cos^2 x + \cos^4 x$ is equal to
2. $\cot 16^\circ \cot 44^\circ + \cot 44^\circ \cot 76^\circ - \cot 76^\circ \cot 16^\circ =$
3. If the straight line $y = mx + c$ ($m > 0$) touches the parabola $y^2 = 8(x + 2)$, then the minimum value taken by c is
4. Normal PO, PA and PB ('O' being the origin) are drawn to $y^2 = 4x$ from $P(h, 0)$. If $\angle AOB = \pi/2$, then area of quadrilateral OAPB is equal to
5. Number of values of m for which the lines $x + y - 1 = 0$, $(m - 1)x + (m^2 - 7)y - 5 = 0$ & $(m - 2)x + (2m - 5)y = 0$ are concurrent ; are

